



Proseminar: Pioneers of Artificial Intelligence

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**From Photons to Phenomenology:
The Scientific Evolution of Fei-Fei Li**

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1 Introduction: The Human Story Beyond Algorithms

"I believe that if we want machines to think, we need to teach them to see."

Prof. Dr. Fei-Fei Li



Fig. 1: Prof. Dr. Fei-Fei Li Portrait [1]

In this day and age, Artificial Intelligence (AI) has become one of the most important technological developments. This journey, ranging from self-driving cars to large language models (LLMs) that can answer questions and engage in conversation, is often described as highly complex in technical and mathematical terms. However, reading the history of this technological revolution solely through complex lines of code or hardware capabilities means overlooking the most important part of the big picture: **"The Data Factor."**

The main purpose of this report is to examine the academic and personal career of Prof. Dr. Fei-Fei Li [1], one of the pioneers whose work in computer vision laid the foundations for today's artificial intelligence. Li's memories and academic work, detailed in her book *The Worlds I See*, show that science is not only done in sterile laboratories isolated from the world. On the contrary, scientific discoveries are inextricably linked to an immigrant's struggle for survival, a child's responsibility to their family, and a researcher's intellectual rebellion against the existing order [1].

Prof. Dr. Fei-Fei Li's career coincides precisely with the transition of artificial intelligence from its 'dark age' (AI Winter II) to its current 'golden age.' Li's story is, in a way, the biography of modern AI. Therefore, this work is not only a portrait of a scientist but also a portrait of technology's human face.

2 The Formation of Scientific Identity

To completely understand Fei-Fei Li's scientific journey, I believe we must analyse the person who most influenced her intellectual development: her sociological background, namely her experience of migration. In the early 1990s, when she was still a young girl, her family's migration from Chengdu, China, to the United States in pursuit of ideals such as freedom and education was not merely a geographical relocation but also a process of identity construction [1].

2.1 The Impact of a Mentor

As she explains in her book in a candid tone, Li's early years in America were far removed from academic brilliance. Weekends in Parsippany, New Jersey, were spent in the dry cleaning shop she ran with her family, filled with the smell of chemicals and the routine

of labelling customers' clothes. A student solving differential equations at school during the week turned into a 'stain remover' expert at work on weekends [1].

For Li, who struggled with language barriers and cultural incompatibility and found it difficult to dream of university due to her family's financial struggles, her maths teacher at Parsippany High School, Mr. Bob Sabella, was vital. Mr. Sabella was not just a teacher to her, but a father figure, and played a huge role in her acceptance to Princeton University's physics department on a full scholarship [1].

2.2 From Physics to The Universe of The Mind

Li's years at Princeton were a search for an intellectual "North Star." Following in the footsteps of her childhood hero, Albert Einstein, she chose to study physics [1].

However, as her undergraduate studies progressed, Li's interest began to shift from the 'outer world' (the universe) to the 'inner world' (the mind) that perceives this world. In this transformation, books such as Erwin Schrödinger's legendary work *What is Life?* and Roger Penrose's *The Emperor's New Mind* served as catalysts [1]. These texts, written by physicists on biology and consciousness, led Li to ask the question: *To understand the universe, must we not first understand the instrument that observes it, namely the brain?*[1]

This intellectual turning point abruptly shifted Li's career from physics to computational neuroscience, which was not very popular at the time [1].

2.3 Summer Research: The Neurological Foundations of Vision

Li's theoretical interest was put into practice during a summer internship at UC Berkeley. This experience marked the beginning of her lifelong passion for the phenomenon of 'vision'. In the laboratory, by analysing electrical signals taken from a cat's visual cortex to the Lateral Geniculate Nucleus (LGN), they succeeded in recreating the black and white pixelated film the cat was seeing at that moment on a computer screen, and this experiment was a fascinating moment for Li [1]. It proved that biological signals (neuronal firings) could be converted into mathematical algorithms. With this experience, Li went to the California Institute of Technology (Caltech) for her doctoral studies and began working with her renowned mentor, Pietro Perona [1].

During her time at Caltech, Li defined the act of 'seeing' not as an engineering problem, but as a learning process. At that time, computer vision experts were trying to write rigid rules to recognise objects. However, inspired by biology, Li and Perona asked the question: *If a child learns about the world, can't a machine learn in the same way?* [1]

This would form the basis of the One-Shot Learning and ImageNet database she would later develop and of the foundation of artificial intelligence: mathematics that mimics biology [1].

3 The Data Paradigm Shift

One of Prof. Dr. Fei-Fei Li's most significant contributions to the history of artificial intelligence was shifting the focus from complex, hand-written algorithms to the quality and quantity of data. This paradigm shift was shaped by work at two opposite ends of the spectrum: first, modelling learning in situations where data is scarce (One-Shot), and then scaling learning by bringing data to massive sizes (ImageNet) [2, 3].

3.1 The Bayesian Mathematics of Learning with Little Data

Before 'Big Data' became popular, the main problem in machine learning was generalisation. Traditional models required thousands of examples to learn a single object. For example, to teach a computer what a 'car' is, you had to show it thousands of car photographs. But humans are different. When a child asks their parent what a chair is and the parent tells them it is a chair, the child can remember it forever. Because he/she had learned it, and this is called 'One-shot Learning'. Li put it this way: *Children are not blank slates (tabula rasa); they use 'prior knowledge' when perceiving the world* [1, 2].

Li, Fergus, and Perona attempted to give machines this human-like ability in a paper published in 2006. Li and her team asked the following questions: *'How can we mathematically model this incredible ability of the human brain in machines? How can a computer learn a new category with very little data?'* [1, 2] In their paper, they proposed a Bayesian method that uses 'prior knowledge'. To understand this, imagine you want to learn what a 'zebra' is. If you don't know what animals are, this is difficult. But if you already know what a horse is, it's easy: a zebra is a striped horse [2].

Li and her team needed a dataset to create this prior knowledge. However, at that time, researchers were using their own narrow datasets (e.g., photos of 5-10 coffee cups). Li decided to create the Caltech-101 dataset to solve this chaos [1]. The method was so primitive that she even mentions it in her book: *Google Images was still new in 2003-2004. I remember my students and I spending hours at the computer manually downloading images and categorising them* [1]. While doing all this at Caltech, Li also continued to be the 'problem solver' for her family's dry cleaning shop in Parsippany [1]. *She recounts moments when her phone would ring while she was working on complex mathematical equations in the lab. It would be her mother on the phone, telling her that the shop's pressing machine had broken down or that she was dealing with an angry customer. Li would leave the lab and head to the car park, trying to calm the customer on the phone while simultaneously translating for her mother in English* [1].

They finally created the Caltech-101 dataset, and based on this logic, Li and her team used general rules from categories the computer already knew (such as general shapes and textures) to enable it to learn new categories much faster. In their work, they used a 'Constellation Model' that represented objects as a probabilistic configuration of parts and

created general rules using different categories such as spotted cats, faces, and aeroplanes as prior knowledge [2].

In my opinion, the use of the ‘Constellation Model’ allowed the system to model the relative positions of features and made it robust against deformations, even with minimal data. This work was important because it showed that ‘learning’ developed over time. However, it was mathematically very complex and required a lot of manual setup, which made it difficult to use for everything [2].

3.2 Redefining the North Star: The ImageNet



Figure 2: **The ImageNet Hierarchy.** A visualization of how ImageNet organizes images according to the semantic structure of WordNet [3].

Li, who returned to Princeton University as an assistant professor in 2007, found her ‘North Star’ in psychologist Irving Biederman’s theory of ‘the human mind’s capacity to recognise 30,000 object categories’ and, together with her team, built the ImageNet project on the backbone of the WordNet structure [1]. This project was launched with the aim of organising the data explosion created by the digital age and creating the critical data source needed to train computer vision algorithms and was born with the goal of organising the massive data explosion brought about by the digital age and meeting the need for quality data required to train computer vision algorithms. Although existing datasets, such as TinyImage, had large scales, such as 80 million images, they were limited by low resolution and high noise levels [3]. Projects such as the ESP Game gamified the labelling process but were insufficient in resolving ambiguities in word meanings. Seeking to fill this gap, Li and her team created an ambitious ontology that aimed to populate the majority of WordNet’s 80,000 concept synsets with an average of 500-1000 clean, full-resolution images for each [1, 3]. As of the 2009 publication of their work, the project had reached a massive structure comprising 5,247 categories grouped under 12 subtrees, such as mammal, vehicle, and furniture, and a total of 3.2 million cleaned images 2. This structure is not just a collection of images, but a dense, semantic hierarchy linked by ‘IS-A’ relationships [3].

Since building such a large-scale database was not possible using traditional methods, the team utilised a mass workforce by using the Amazon Mechanical Turk (AMT) platform, which was mentioned by chance by Li’s master’s student, Min [1]. This process began with searching for WordNet synonyms in search engines using multilingual queries, followed by the verification of the collected candidate images by AMT users [1, 3]. To prevent human error and data contamination, each image was rated by multiple users, and a dynamic confidence score table algorithm was developed to determine the required number of votes based on the difficulty level of the categories (e.g., the general ‘cat’ category versus the specific ‘Burmese cat’ distinction) [3]. As a result of this process, an average labeling accuracy of 99.7% was achieved even at the lowest levels of the hierarchy. Furthermore, the diversity of the images was verified by analyzing the average lossless JPG file sizes of the images in the categories, and it was determined that the fuzzy average images also exhibited high diversity [3].

ImageNet’s technical superiority over its competitors was demonstrated by three fundamental application experiments presented in the paper. In the first object recognition experiment, using the Naive Bayesian Nearest Neighbour (NBNN) method, it was shown that ImageNet significantly improved classification performance on clean, full-resolution data compared to noisy data [3]. In the second experiment, it was observed that the “Tree-max classifier” method, which utilizes the hierarchical structure of the dataset, was more successful than independent classifiers, and that objects at lower levels of the tree were easier to classify due to visual consistency [3]. Finally, experiments conducted across 22 categories positioned ImageNet as a challenging benchmark for object detection and localization algorithms [3].

In my opinion, integrating the WordNet hierarchy into visual data enabled the system to learn not only the object but also the semantic relationships between classes, as seen in the “Tree-Max Classifier” experiment. This transformed the data from a flat list of labels into a much deeper structure. However, as the images were entirely retrieved from search engines (keyword queries) [3], there is a risk that the set reflects the “bias” of search engines rather than the natural world distribution; for example, typing ‘engineer’ brings up images of men, while typing “nurse” brings up images of women... This could affect the model’s generalisation success in real life.

3.3 The Turning Point and the Age of Deep Learning

When the ImageNet dataset was presented to the academic community in 2009, it was initially met with skepticism and dismissed due to the algorithm-centric paradigm of the time [1]. Fei-Fei Li launched the ILSVRC competition to prove the value of the data. This competition challenged researchers: *“Your algorithms work on 100 cat images, but what about 1,000 cat images? What about 1 million images?”* [1].

For the first two years (2010 and 2011), results using traditional methods (“hand-crafted features”) showed very little progress. Error rates remained stuck around 25% [1]. Everyone began to think that computer vision had reached its limits. That is, until 2012, the moment of the “big bang”.

The real turning point/paradigm shift in computer vision occurred in 2012 with AlexNet, developed by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton [1]. This model dramatically reduced error rates (from 26% to 15.3%) through the synthesis of Convolutional Neural Networks (CNNs), GPU processing power, and large-scale data, namely ImageNet [1, 4]. This success officially ended the so-called “AI Winter” and is one of the most important events that launched the modern era of deep learning, historically proving the scientific validity of the data-driven approach [1].

4 Connecting Vision and Language

The ImageNet project and the AlexNet revolution in 2012 largely solved the problem of “Object Recognition” in artificial intelligence, enabling it to assign a single label (e.g., “Cat”) to images. However, for Li and her team, this was only a superficial imitation of human vision. When humans look at the world, they see relationships and stories, not isolated labels. As Li illustrates in her book, using her son Leo’s language development, the cognitive process involves not just pointing, but complex descriptions like “The car is going fast.” [1]

4.1 Beyond Words and Dense Captioning

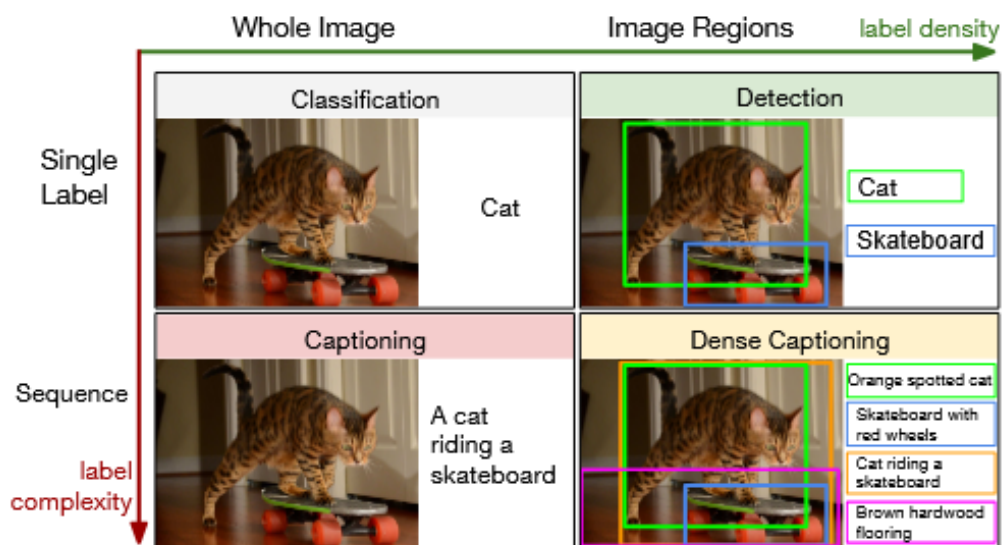


Figure 3: **From Classification to Dense Captioning.** The model localizes and describes regions within a scene [5].

This project is a milestone in the literature because it combines two different AI disciplines into a single end-to-end architecture called the Fully Convolutional Localization Network (FCLN) [5]. The model works as follows: Eyes (CNN): The system scans the image using the VGG-16 architecture and extracts visual features in a single forward pass [5]. This is critical for efficiency because it prevents repeated processing for each region.

Focus (Localization Layer): This is the heart of the model. The team used Bilinear Interpolation to enable the network to learn regions of interest in the image in a derivable manner. This allows the model to learn where to look on its own, without requiring an external region proposal algorithm [5].

Mouth (RNN): For each detected region, visual features are sent to an LSTM (Long Short-Term Memory) language model. The RNN processes these features to generate grammatically correct sentences such as “The black cat is sitting on the grass” or “The man is wearing a red hat” 3 [5].

As a result, the model can detect hundreds of different regions in a single photo and describe not only the elephant itself, but also details such as “the elephant’s trunk” or “the elephant’s foot” separately [5].

From my perspective, using the Bilinear Interpolation mechanism instead of RoI Pooling enabled the system to be trained end-to-end for the first time by making the coordinates differentiable. This move freed the model from external region proposal algorithms and made it much faster. However, the continued reliance on an intuitive pruning method like Non-Maximum Suppression (NMS) during testing and the frequent occurrence of semantic repetition in the outputs indicate that the architecture does not fully ensure contextual integrity [5].

4.2 Visual Genome

Although the ImageNet project and the Deep Learning revolution of 2012 largely solved the problem of “object recognition” in artificial intelligence, creating a visual vocabulary, for Li and her team, this was only a superficial imitation of human vision; because, as Li expressed using the astrophysical analogy of “Dark Matter,” the invisible element that constitutes the bulk of the visual world is “relationships” [1]. Because ImageNet only contained nouns (objects) and left out verbs and prepositions (interactions), the Visual Genome project, which aimed to transition from the perceptual to the cognitive level, was designed as the logical successor to ImageNet and, unlike traditional sets such as MS COCO, to transform the visual into a dense data network [6].

If ImageNet is the vocabulary of artificial intelligence, Visual Genome is the grammar that makes sense of its words. At the heart of the project is the Scene Graph model, which models visual data as a directed graph. This model enables artificial intelligence to answer questions beyond ‘what is there?’ to ‘what is happening there and why?’ (VQA) [6].

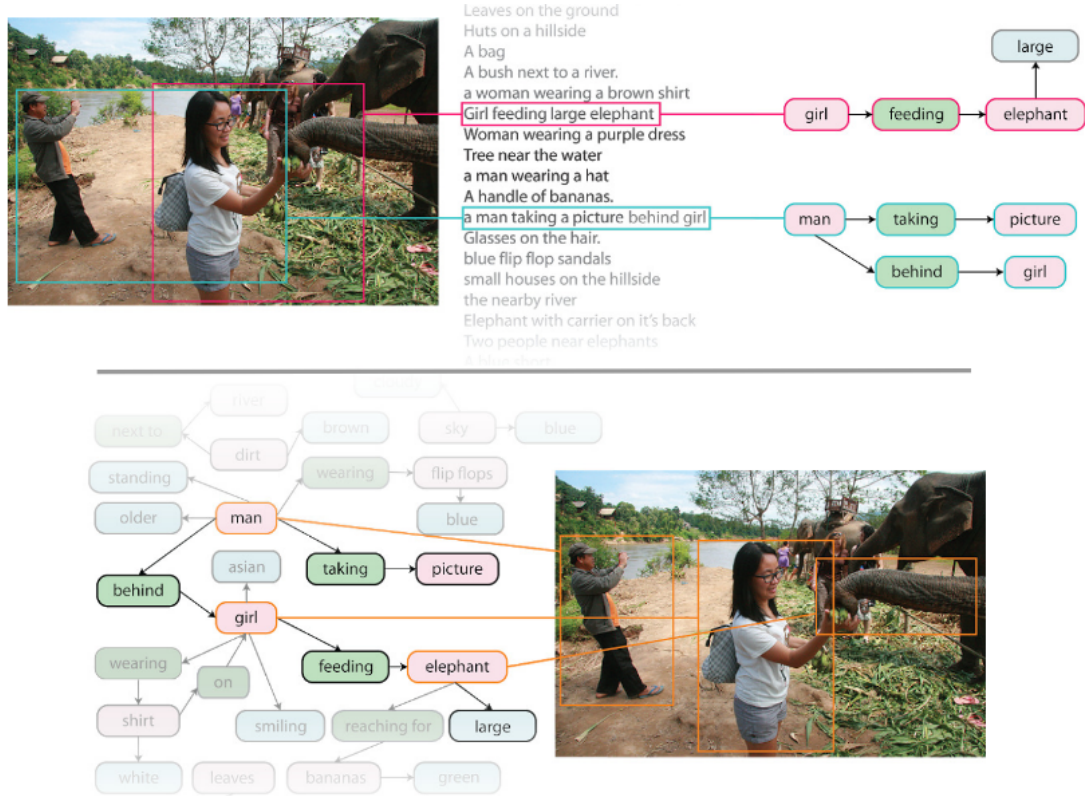


Figure 4: **Visual Genome Scene Graph.** An illustration of how visual data is structured as objects and relationships [6].

This structure contains concrete ‘Objects’ (Nodes) defined by Bounding Boxes and Word-Net IDs, ‘Attributes’ that characterise these objects (Nodes), and critical ‘Relationships’ (Edges) that determine the meaning of the event [6]. Created by the Stanford team led by Ranjay Krishna using Amazon Mechanical Turk, this massive knowledge base contains 3.8 million objects, 2.8 million features, 2.3 million relationships, 1.7 million question-answer pairs, and a total of 5.4 million region annotations, averaging 50 annotations per image [6]. Thanks to this data, artificial intelligence can now distinguish semantic differences in scenes with similar pixels (e.g., a man riding a horse versus a man standing next to a horse) [6]. An academic milestone, the Visual Genome enables the transition from pixels to semantics, developing the ability for ‘Relational Reasoning,’ but it also introduces limitations, such as noise from crowd-sourced data and the scarcity of rare relationships (long-tail distribution) [6]. Ultimately, Fei-Fei Li challenged researchers tackle more complex, difficult problems with this project. She also demonstrated the importance of the qualitative depth of ‘Big Data’ and laid the foundation for today’s multimodal artificial intelligence models [1].

In my view, Scene Graph modeling, by extracting images from a flat pile of labels and transforming them into a structured information network, gave artificial intelligence the ability to ‘reason relationally’ for the first time. However, the long-tail distribution of relationships in the dataset and the noise introduced by crowd-sourced production make it

difficult for the model to learn rare interactions, posing a risk of limiting its generalization success in complex real-world scenarios [6].

5 Human-Centered AI

The journey of artificial intelligence that teaches machines to “see” with ImageNet has evolved through stages of understanding relationships between objects (Visual Genome) and actions over time (Action Recognition), ultimately transforming into a concrete care tool with Ambient Intelligence, Fei-Fei Li’s most personal project. Li’s mother’s chronic heart condition and the helplessness she experienced during hospital stays were the driving force behind this transformation [1]. Li defined the moments when patients are left alone, away from the supervision of doctors and nurses, as the “Dark Spaces of Healthcare” and decided to use technology to fill this void [1]. To address this lack of observation, one of the biggest causes behind up to 400,000 preventable deaths each year in the US, she collaborated with Stanford Medical School to develop systems that make physical environments “responsive” [1, 7].

5.1 Ambient Intelligence in Healthcare

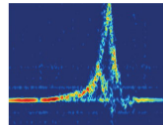
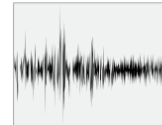
	Camera	Depth sensor	Thermal sensor	Radio sensor	Acoustic sensor
Sensory information	RGB, colour, video	Lidar	Infrared	Radar, Wi-Fi	Microphone
Function	Measures colour (visible light)	Measures distance to objects	Measures surface temperature	Estimates distance and velocity	Measures air pressure waves (sound)
Sampling rate	30 Hz (1,920 × 1,080)	30 Hz (1,280 × 720)	10 Hz (640 × 480)	800 Hz	44.1 kHz
Bit depth	24 bits	16 bits	16 bits	32 bits	16 bits
Uses	Object recognition, person detection	3D object detection, robotic navigation	Night vision, equipment safety	Motion detection, object detection	Speech recognition, event detection
Data visualization					

Figure 5: **Privacy-Preserving Ambient Intelligence.** Using depth sensors to monitor behavior without recording faces [7].

The technical architecture of this project aims to eliminate the limitations of wearable technologies, such as charging and comfort, by placing sensors in the environment rather than on the person. As traditional RGB cameras pose a threat to data privacy and ethical concerns, depth and thermal sensors have been used in this project [7]. These are used to analyse only the body silhouettes and movement skeletons of patients, not their identities 5. This system detects critical situations such as the patient’s risk of falling, hand hygiene violations, or irregular breathing, and enables notification to healthcare personnel [7]. Naturally, patient privacy is protected while all this is being done [7].

Artificial Intelligence is reaching a threshold where it can interpret not only the digital world but also the physical world through Ambient Intelligence [7]. Through continuous, passive monitoring, we can eliminate the Hawthorne Effect and related observer effects, enabling us to collect the most natural clinical data (Ground Truth) [7]. However, monitoring patients at their most vulnerable moments immediately brings to mind the concept of ‘Big Brother’ from George Orwell’s 1984, and it also brings with it serious sociological and ethical risks, such as the privacy paradox. Li’s work is the strongest evidence that technology requires not only technical success but also a ‘human-centred’ and ethical design process [7].

To my mind, Ambient Intelligence is the stage where artificial intelligence, which learned to see with ImageNet, has now learned to ‘look’ and ‘listen’. Unlike wearable technologies, taking technology off the human body and placing it on the wall is a wonderful example of the ‘Calm Tech’ philosophy. However, as poetic as the metaphor of ‘Lighting Up Dark Areas’ may be, success lies not in the intelligence of the algorithm, but in the guarantee of privacy it offers [7].

5.2 The Future: Spatial Intelligence

Li’s current research is about “Spatial Intelligence.” She thinks today’s popular AI models like ChatGPT are incomplete. She calls them “wordsmiths in the dark” because they can write beautiful poetry but cannot make a cup of coffee [8].

She imagines a future where AI has “world models.” This means AI will understand 3D geometry and physics. Unlike a chatbot, a spatially intelligent robot could walk into a messy kitchen, find an apple, wash it, and put it on a plate. This vision brings Li’s career full circle: from identifying static images to creating AI agents that can act in the physical world [8].

6 Summary

This report traces the scientific development of Prof. Dr. Li Fei-Fei, demonstrating how the focus of Artificial Intelligence has shifted from algorithms to data. In 2006, as discussed in the section on ‘One-Shot Learning,’ which was developed based on how a child learns, this field was unfortunately stuck, attempting to learn from very few examples. But Li proved that AI needs a large amount of experience and data to learn like a human being by creating a dataset of millions of images, rather than the Caltech-101 dataset, which she compiled by downloading images one by one from the internet. This work effectively launched the modern era of deep learning.

However, recognising objects was only the first step. As shown in the sections on Dense Captions and Visual Genome, Li realised that ‘seeing’ was not just about naming objects, but also about understanding relationships and telling stories. In her work with her team, she enabled her machines to understand computer vision through language, allowing them to recognise not just the ‘motor’ and the “woman” separately in an image, but rather the ‘woman riding the motor’. This transition was a crucial step forward in making artificial intelligence more useful.

Furthermore, her recent focus on Ambient Intelligence following her mother’s hospitalisation demonstrates that technology should also serve human needs in hospitals. The use of sensors to monitor hygiene or patient safety in hospitals is a powerful example of ‘Human-Centred AI’. However, as noted in the report, this unfortunately raises serious ethical issues regarding privacy. The decision to use depth sensors instead of cameras is a smart technical solution, but the fine line between security and surveillance will remain an important issue for future researchers.

Moreover, Li’s current vision for spatial intelligence suggests that the next major leap will be physical interaction. Moving from ‘words’ (language models) to ‘worlds’ (robotic agents) means that artificial intelligence will soon need to understand physics and 3D space. I believe this is the most exciting direction, as it suggests AI will move beyond digital screens and into our physical reality.

Finally, Prof. Dr. Li Fei-Fei’s career teaches us that intelligence is not just about data, but also about interacting with the complex and fragmented world we live in. And all her projects show how human life is, highlighting the difficulties faced by a migrant woman/girl.

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